

GLOBAL MACRO TRADING & RISK ENGINE

Independent FICC & Equities research project | Executed back-test: 2021–2025 | Market snapshot: September 2026

Executive summary

This project tests a deliberately simple, lagged macro regime framework for a multi-asset portfolio. Each year's allocation is determined only from information available through the previous year-end, avoiding look-ahead bias. The model combines monetary-policy direction, energy/inflation pressure and equity-market momentum to classify the environment as Risk-on, Balanced or Defensive.

Headline result: the strategy compounded by 50.7% from 2021–2025 (8.5% CAGR), with 11.5% annual volatility and a 7.3% maximum drawdown. The S&P 500 price index compounded by 82.3% (12.8% CAGR) over the same period, with a 19.4% maximum drawdown. The strategy therefore traded some upside for substantially lower downside during the 2022 sell-off.

The project is designed as market research rather than an investable strategy. Brent is treated as a commodity-price proxy, cash is approximated with lagged Fed Funds, and the S&P 500 series excludes dividends.

Strategy cumulative	Strategy CAGR	Volatility	Max drawdown	Sharpe
50.7%	8.5%	11.5%	(7.3%)	0.54x

1. Research question

Can a rules-based macro framework identify changes in market regime early enough to improve downside control without relying on forecasts of individual securities?

2. Model architecture

Signal	Rule	Score
Monetary stance	Prior year-end Fed Funds vs previous year-end; <0.9x = easing; >1.1x = tightening	+1 / 0 / -1
Energy / inflation	Prior year-end Brent vs previous year-end; <0.9x = disinflation; >1.1x = inflation pressure	+1 / 0 / -1
Risk momentum	Prior year-end S&P 500 above/below previous year-end	+1 / -1
Composite	Sum of the three signals	-3 to +3

Target weights used by the model:

Regime	S&P 500	Brent	Cash
Risk-on	65%	15%	20%
Balanced	45%	15%	40%
Defensive	20%	15%	65%

3. Executed back-test

Year	Regime	Strategy	S&P 500	Strategy cumulative	S&P cumulative	Strategy DD
2021	Risk-on	24.7%	26.9%	1.247	1.269	0.0%
2022	Balanced	-7.3%	-19.4%	1.156	1.022	-7.3%

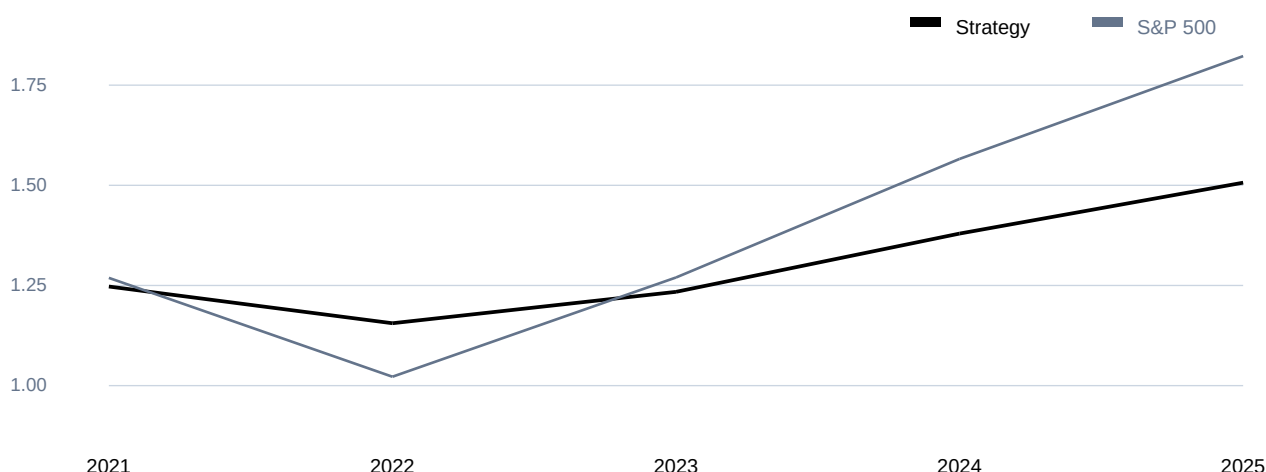
Year	Regime	Strategy	S&P 500	Strategy cumulative	S&P cumulative	Strategy DD
2023	Defensive	7.2%	24.2%	1.239	1.270	-0.6%
2024	Balanced	11.8%	23.3%	1.385	1.566	0.0%
2025	Risk-on	9.2%	16.4%	1.513	1.823	0.0%

The model's strongest evidence of value is 2022: the Balanced regime held the strategy's annual loss to 7.3% versus 19.4% for the S&P 500. The cost was lower participation in the strong 2023 and 2024 equity rebounds, demonstrating a central trade-off between protection and upside capture.

4. Risk analysis

The strategy's 11.5% annual volatility is lower than the variability implied by the S&P-only path in this five-year sample. Its 7.3% maximum drawdown occurred in 2022 and was recovered before the end of the test period. The 0.54x Sharpe ratio is calculated using the lagged Fed Funds proxy as the annual risk-free rate.

Cumulative growth of £1



5. 2026 market snapshot and model output

Indicator	Latest	Period	Change / note
S&P 500	7,656.98	11 Sep 2026	+11.9% vs 31 Dec 2025
Fed Funds	3.63%	Aug 2026	-2.4% vs Dec 2025 average
Brent	\$83.73/bbl	Jul 2026	+35.5% vs Dec 2025 average
GBP/USD	1.3547	Aug 2026	+1.2% vs Dec 2025 average
US 10Y	4.95%	10 Sep 2026	Elevated rates backdrop

Model output: Composite score = 0 → Balanced. Target weights: 45% S&P 500 / 15% Brent / 40% cash. The model's interpretation is that positive equity momentum is offset by materially higher energy prices, while the policy-rate signal is neutral under the $\pm 10\%$ threshold.

6. Stress testing

Scenario	S&P shock	Brent shock	Portfolio shock
Risk-off sell-off	-15%	-20%	-9.8%

Scenario	S&P shock	Brent shock	Portfolio shock
Inflation shock	−10%	+30%	−0.0%
Rates shock	−8%	+5%	−2.9%
Severe stagflation	−20%	+40%	−3.0%
2022-style equity drawdown	−19%	+9%	−7.2%

The stress tests show the portfolio is most dependent on the cash sleeve for broad risk-off protection, while the Brent allocation acts as a partial hedge when inflation shocks push energy prices higher.

7. What the model got wrong

The framework is intentionally not optimised to hindsight. The clearest weakness was 2023: the prior-year defensive signal reduced participation in a 24.2% S&P rebound. The strategy also lagged in 2024 because the Balanced regime capped equity exposure while equities rallied. These misses are useful: they demonstrate that a macro regime model can reduce drawdowns while still suffering from regime-timing error.

8. How this demonstrates FICC & Equities skills

Market research: links inflation, rates, energy and risk sentiment to asset prices.

Analytics: converts raw data into signals, portfolio weights and performance attribution.

Risk management: measures volatility, drawdown, correlations and stress losses.

Decision-making: forms explicit positioning rules before observing subsequent returns.

Communication: translates the model into a concise market view and trade-off analysis.

9. Limitations & integrity statement

This is an educational research project, not investment advice. The back-test has only five annual observations; Brent is represented by monthly average spot-price data rather than futures P&L; the S&P 500 is a price index without dividends; cash uses a lagged Fed Funds proxy; and the 2026 snapshot uses different latest-available dates across series. The model is deliberately simple and should not be represented as evidence of a tradable alpha strategy.

10. Data sources

FRED — S&P 500: <https://fred.stlouisfed.org/series/SP500>

FRED — Federal Funds Rate: <https://fred.stlouisfed.org/series/FEDFUNDS>

FRED — Brent crude: <https://fred.stlouisfed.org/series/POILBREUSDM>

FRED — GBP/USD: <https://fred.stlouisfed.org/series/DEXUSUK>

FRED — 10-Year Treasury: <https://fred.stlouisfed.org/series/DGS10>

Bank of England — Bank Rate history/data: <https://www.bankofengland.co.uk/boeapps/database/Bank-Rate.asp>